



OPERANT
— NETWORKS —

NDNComm 2020

*Improving Existing Software Applications with a Practical and
Secure NDN Publish/Subscribe Transport*

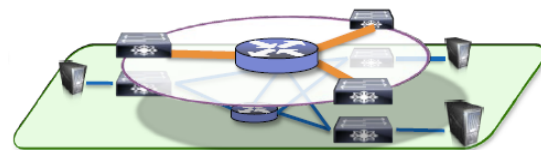
- Randy King, CTO Operant Networks

Challenges in Commercializing NDN

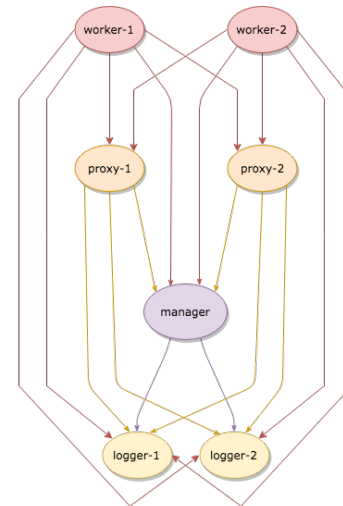
- When Operant introduces NDN to customers, we have to tell them it works 'differently'...but now we've highlighted a barrier to adoption
 - For customers to adopt a new technology they must be strongly motivated; it can't just be 10% better
 - What are the possible motivators for NDN over existing solutions?
 - Security <- *Important (even trending) but hard to quantify and value*
 - Multiparty communications <- *Not common yet...on the horizon*
 - Multiple physical layers in series or parallel <- *Many workarounds*
 - Caching & data throughput <- *Large customers have solutions in place*
 - Few 'lockout' applications; we try to find ones with multiple motivators
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Intrusion Detection Systems

- **Firewalls** are the most common cybersecurity tool protecting networks
 - Inspect packet *payloads* traversing site perimeters
 - Growth of encrypted traffic makes deep packet inspection difficult
 - Sophisticated hackers find a way through eventually
- **Intrusion Detection Systems (IDS)** are an alternative
 - Observe TCP/IP packet *headers* and traffic flow pattern variations
 - Valuable since hackers generally 'lurk' before attacking
 - Widely distributed traffic sensors also reveal internal flow anomalies
- **IDS must be more secure than customer network**
 - USAF and DOE agreed and funded an NDN 'stealth overlay' distributed IDS



- Customers don't generally buy networking, they buy full solutions
- Many existing cybersecurity tool solution providers and a robust open source community
- To enter the market, Operant leveraged the Zeek IDS
- Zeek is a distributed sensor-based system that preprocesses observed network traffic flows and creates logs and alerts for further analysis in a SIEM
- Zeek has a broker-based pub/sub communications framework using passwordless SSH TCP/IP logins
- Operant has replaced that framework with an *end-to-end secure* NDN-based *broker-less* pub/sub transport



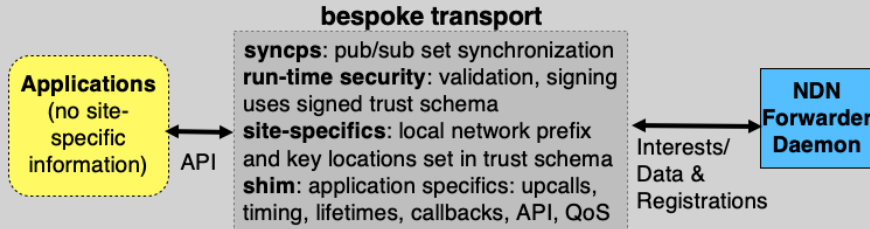
Securing Pub/Sub in a TCP/IP World

- MQTT is a lightweight TCP/IP protocol for resource-constrained IoT devices and non-ideal networks
 - Requires a Broker as a relay between publishers and subscribers
 - Broker must know all node IP addresses and subscriptions
 - Broker is a target for cyber-attack and a single point of failure
 - No security; nodes are not authenticated and data is not encrypted
- DDS is decentralized; publishers and subscribers communicate as peers
 - Store all node addresses and subscribed topics within every node; requiring an ongoing dynamic discovery process
 - *'DDS was, however, not designed to be used over disadvantaged networks...this standard is not efficient enough'*
 - STO TECHNICAL REPORT 'SOA Challenges for Real-Time and Disadvantaged Grids'

Pollere Sentinel Bespoke Transport

- Based on prior DNMP

Decompose: communications models are implemented by a bespoke transport containing the sync and application-specific functions



security: run-time validation, signing, and publication construction uses a signed compiled binary of the application's trust schema

shim: provides application specificity using library or template functional modules including publication expiration, API, sending priority and delivery QoS (e.g. at most once, at least once)

bespoke transport is just a name for the collection of modules that implement an NDN user-space information transport, not a new architectural feature

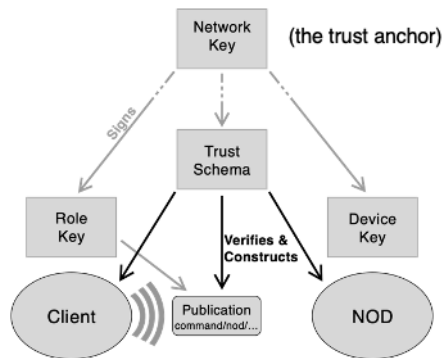
- A message-based pub/sub with a different bespoke transport as expressed in the shim and trust schema
- Uses the same syncps and some software modules: a new transport doesn't have to start over again
- We followed an MQTT-like API to increase application opportunities

- The trust schema applies to publications, not the outer Data packet
- Human readable trust rules are compiled into a trust scheme binary

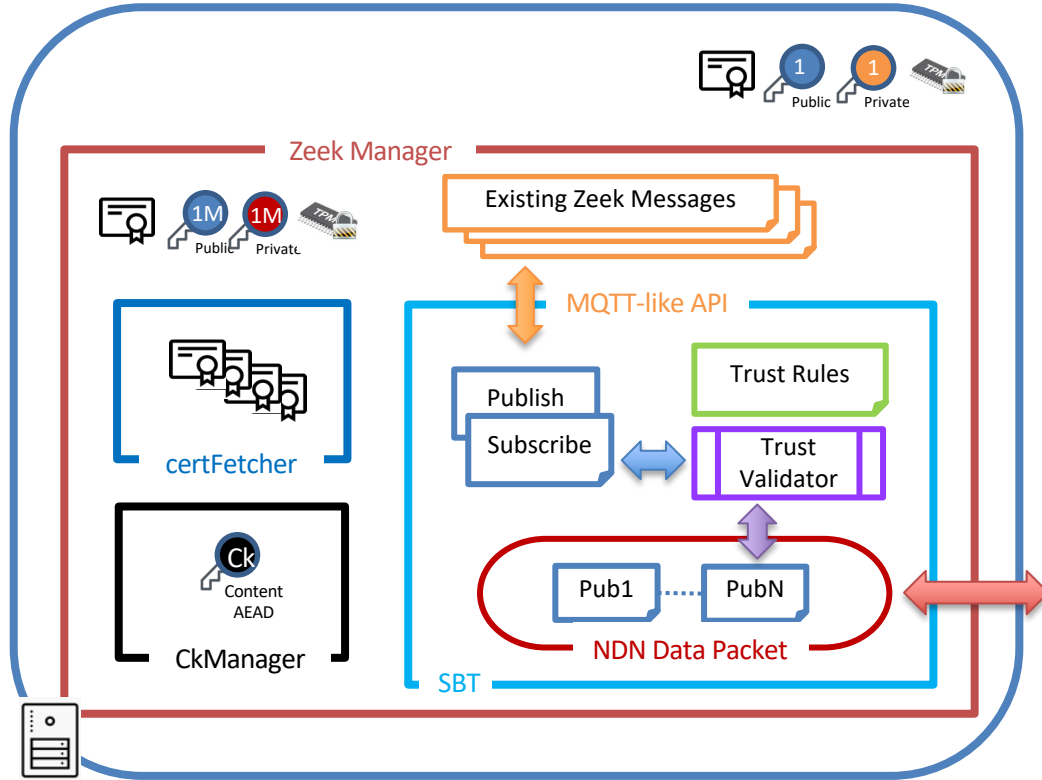
```
logsnetTraff: #sbtPub {
  target: "manager"
  messageType: "logs"
  collection: "CaptureLoss::LOG" | "Conn::LOG" |
  "PacketFilter::LOG" | "Reporter::LOG" | "SSH::LOG"
}
logsnetTraff <= workerCert
```

A New Approach: Versatile Security Toolkit (VerSec)

- A language and compiler for trust rules that checks entire schema, then outputs it in a *signable* compact binary form as key
- Run-time security object, *schemer*, for validation and Data name construction, also allows applications to reference Name components by names!



Sentinel Operation



- *Publications* are signed by the Zeek entity private key and optionally encrypted via RSA
- If they meet the trust rules, SBT will build the publications and add to a *Data packet*, encrypt/authenticate with Ck, and sync via UDP broadcast
- Received *Data packets* are decrypted/authenticated with Ck
- Publications are extracted; if they meet specific trust rules for the topic and role, they are passed to *Subscribers*

- What is ndn-ind?
 - An LGPL open source library extending ndn-cpp for *industrial* uses
 - Security aspects under review by National Labs
 - Will be standardized and change controlled per industry needs
 - Changes for Sentinel
 - Scheduler from ndn-cxx used to schedule sending sync interests
 - *std::chrono* for time stamp and duration to avoid confusion of API
 - AEAD algorithm ChaCha20-Poly1305 which requires OpenSSL 1.1
 - In NAC, let the access manager create a group AEAD key periodically refreshed and fetched by the publisher for encrypting
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- Sentinel IDS has been implemented with a secure broker-less NDN pub/sub transport
- Customer-specific SIEMs have been developed for Air Force and DOE Solar
- This demonstrates retrofit of existing TCP/IP applications with a commercially viable effort level

Next:

- Utility Communications
- MQTT-like apps: brokerless & secure

